C4.1 Populations and communities

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c4-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 548–577 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Populations and sampling

C4.1.1 Populations

p. 548

A **population** is a group of organisms of the same species living in the same area at the same time. Its members normally interbreed. One population is separated from another of the same species by **reproductive isolation**, for example by distance or a barrier.

C4.1.2 Why we sample

pp. 549–550

Counting every individual is usually impossible, so we count a sample and scale up to an estimate. The sample must be **random**, so that every part of the area has an equal chance of being sampled and the result is not biased.

C4.1.3 Random quadrats

pp. 550–551

For sessile organisms, place quadrats at random coordinates, count the individuals in each and find the mean. Population size = mean per quadrat × total area ÷ area of one quadrat. The **standard deviation** of the counts shows how much they vary, and so how evenly the organism is spread.

C4.1.4 Capture-mark-release-recapture

pp. 555–556

For motile organisms, capture and mark a sample (M), release them, then capture a second sample (N) and count the marked ones (R). The **Lincoln index** gives population size = (M × N) ÷ R. It assumes that marks are not lost, marked animals mix fully, and there are no births, deaths or migration between samples.

Population growth

C4.1.5 Carrying capacity

p. 557

The **carrying capacity** is the maximum population size that an environment can support. It is set by limited resources, such as food, water, territory and nesting sites for animals, and light, water and minerals for plants.

C4.1.6 Density-dependent control

pp. 557–558

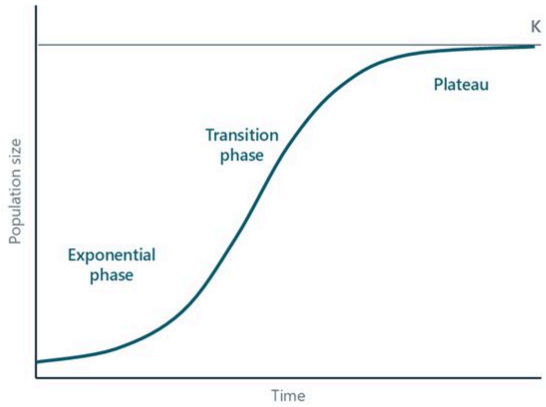
Competition, predation and the spread of pathogens and pests all increase as a population becomes denser. Above the carrying capacity they raise deaths and reduce births, and below it they relax. This **negative feedback** keeps the population near the carrying capacity. Density-independent factors, such as floods or fires, can change numbers suddenly but do not do this.

C4.1.7 The sigmoid growth curve

pp. 558–561

The sigmoid population growth curve

C4.1.7



Exponential: plenty of resources.

Transition: density-dependent factors increase.

Plateau: births = deaths at K.

Example
Bearded vultures released in the Alps since 1986 are now growing fast: early exponential phase.

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In the **exponential phase**, resources are plentiful and limiting factors are weak. In the **transition phase**, density-dependent factors increase and growth slows. At the **plateau**, births and deaths balance at the carrying capacity. The return of the bearded vulture to the Swiss Alps is a growing population.

Watch out: a lag phase is not expected as part of this curve.

C4.1.8 Modelling growth with duckweed

p. 562

Count the fronds of duckweed (*Lemna*) in a container every day or two until the surface is covered. Plot the counts on a log scale against time on a normal scale: exponential growth gives a straight line, and the line curves away as growth slows.

Communities and relationships

C4.1.9 Communities

pp. 562–563

A **community** is all the interacting populations in an area, including plants, animals, fungi and bacteria. A community and its abiotic environment form an ecosystem.

C4.1.10 Competition and cooperation within a species

pp. 563–564

Members of a species have the same needs, so they compete for food, light, territory and mates, as when wildebeest males fight over territory. They also cooperate: hyenas hunt in groups, and leafcutter ant workers gather leaves to grow the fungus that feeds the colony.

C4.1.11 Relationships between species

pp. 564–565

Herbivory (bison and grass), **predation** (lynx and hare), **interspecific competition** (lions and leopards), **mutualism** (Rhizobium and clover), **parasitism** (ticks and deer) and **pathogenicity** (chytrid fungus in the evergreen robber frog).

C4.1.12 Mutualism

pp. 566–568

In legume root nodules, *Rhizobium* fixes nitrogen for the plant and receives sugars. Orchid seeds have no food store, so mycorrhizal fungi supply them with sugars and minerals; adult orchids later supply carbon compounds. In hard corals, zooxanthellae supply products of photosynthesis and receive shelter in sunlit water, carbon dioxide and nutrients.

Competition between species

C4.1.13 Invasive and endemic species

pp. 568–569

An invasive species, introduced by humans, often lacks its natural predators and pathogens, and may take a shared resource more effectively. Grey squirrels outcompete red squirrels for seeds in Britain. Local examples include the nutria, Himalayan balsam and Japanese knotweed.

C4.1.14 Testing for competition

pp. 569–571

Approaches include laboratory experiments, field observations by random sampling, and field manipulation by removing one species, as Connell did with barnacles. If a species does better without the other, competition is indicated but not proven.

C4.1.15 The chi-squared test for association

pp. 572–574

Calculating chi-squared

C4.1.15

Expected	Bracken +	Bracken -
Ling +	30	30
Ling -	20	20

1. Expected = row × column total / grand total
2. Sum $(O - E)^2 / E = 4.8 + 4.8 + 7.2 + 7.2 = 24.0$
3. df = 1; critical value 3.84 (p = 0.05)

Conclusion

24.0 > 3.84: reject H0. A negative association: they co-occur less than expected. Could be competition, allelopathy or different soil needs: the test shows association, not cause.

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Record the presence or absence of two species in many random quadrats. Expected value = row total × column total ÷ grand total. Chi-squared is the sum of $(O - E)^2 \div E$. For a 2 × 2 table there is 1 degree of freedom, and the critical value at p = 0.05 is 3.84. A larger value means the species are associated.

Control of populations

C4.1.16 Predators and prey

pp. 574–575

When snowshoe hares are plentiful, lynx have more food and increase. More lynx eat more hares, so hares fall, and then lynx fall too. The two populations cycle, with the predator lagging behind: predation is density-dependent control.

C4.1.17 Top-down and bottom-up control

pp. 575–576

In **top-down** control, a higher trophic level limits the one below, by predation or herbivory. In **bottom-up** control, the supply of nutrients or producers limits the levels above. Both are possible, but one usually dominates in a community.

C4.1.18 Allelopathy and antibiotics

pp. 576–577

In **allelopathy**, a plant such as bracken or a walnut tree releases chemicals that stop other plants germinating or growing. Fungi such as *Penicillium* secrete **antibiotics** that kill bacteria. In both, a chemical released into the environment deters competitors.

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